Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Conley, Catherine A Ph.D.		
Project Title:	Genomic Analysis of the Hypergravity Response in Caernorhabditis Elegans		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cconley@mail.arc.nasa.gov	**Fax:**	FY 650-604-3954
PI Organization Type:	NASA CENTER	**Phone:**	650-604-0234
Organization Name:	NASA - AMES Research Center		
PI Address 1:	SLR		
PI Address 2:	M/S 239-11		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:	NOTE (June 2016): Dr. Conley is currently at NASA Headquarters		
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	10/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Understanding responses to altered gravity is a major scientific objective of NASA Fundamental Biology. We propose a genomic approach to characterize the responses of the nematode worm Caenorhabditis elegans during chronic hypergravity (hyperG) exposure. The long-term goal of this research is to understand gravity-detecting and gravity-responding genetic pathways in C. elegans at the molecular level. These pathways are likely to be evolutionarily conserved in animals, so understanding the molecular pathways in worms should greatly assist the development of better countermeasures for long-duration spaceflight in humans. Whole-genome microarray expression data from worms exposed to 10xG for four days indicate that only a few hundred genes are detectably altered, out of ~19,000 total. To obtain a complete picture of the hyperG response in C. elegans, we will analyze cultures treated at different levels of hyperG for a series of timepoints. One neurotransmitter gene was altered, that has been reported to regulate behavior in lab-grown worms. Worms videotaped during hyperG treatment and recovery show alterations in behavior that correlate		

Task Description:	with reports of behavioral analyses in neurotransmitter-mutant worms. We will quantitate alterations in behavior observed during hyperG treatment and recovery in both mutant and wild-type worms, to establish whether this neurotransmitter is a 'master control gene' regulating behavioral responses to hyperG. The global profile of expression at hyperG is consistent with a developmental delay, although the expression of stress-response genes is not significantly altered. We will characterize the observed developmental delay by analyzing staged cultures. Based on the data obtained from whole-genome microarray analyses, we will construct molecular models of genetic pathways that potentially respond to hyperG. Models will be tested by analyzing worms that are mutant or transgenic for genes showing altered expression. These will be the first molecular models describing hyperG-responsive genetic pathways generated for any system, and should greatly improve our understanding of biological responses to altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research funded by this grant is already being used for education, to stimulate students' and the public's interest in science and technology, through postdoc training, lectures at universities and museums, and soon on the NASA explores web site. The long-term expected benefits of this research include the identification of previously unrecognized gravity-responsive genes, which may be novel targets for countermeasures to improve the health of astronauts and patients confined to bedrest.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 04/15/2008)